

Building Technology

Basic Mathematics I

Learner Notes

Number operations, fractions, decimals, percentages, ratios, and construction measurements.

Learning outcome	Focus
Perform number operations	Applied mathematics in workplace and training contexts.
Apply fractions and percentages	Applied mathematics in workplace and training contexts.
Use ratios in building works	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Building Technology. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

$$\text{Percentage} = \frac{\text{Part}}{\text{Whole}} \times 100\%$$

$$\text{Ratio} = a : b$$

$$P = 2(l + w)$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

4. Practice Questions

1. Calculate percentages in wastage allowance.

2. Determine mortar mix quantities.
3. Find perimeter of a rectangular building plot.